Highway Engineering

- INS3441 -

PROBLEM SESSION 4



Question – 1

An highway is designed as two-way double lane and its design speed is 80 km/h. The data about the highway is provided below.

a) Find all components of the curve (R, D, t, b).

b) By considering that at most half of the centrifugal force can be compensated by the friction, draw the constant centerline superelevation diagram for this curve and show all necessary measurements (km and superelevation values).

c) The red elevation (grade elevation) of the point Φ is given as 85 m. Calculate the elevations of inner and outer edges.

Project Data	Project Criteria
$\Delta = 32.71°$	Lane width = 3.5 m ; Shoulder width = 2.5 m
$\Phi_{km} = 0 + 315^{00}$	Roof slope = %2 ; Max. superelevation = %8
$\mathbb{F}_{km} = 0 + 452^{02}$	Theoretical superelevation = $0.786*(V^2/R)$

a) Find all components of the curve (R, D, t, b).

$$D = \text{Ғ}_{(km)} - \text{Ф}_{(km)} = [\ 0 + 452^{02}\ ] - [\ 0 + 315^{00}\ ] = 137.02\ \text{m}$$

$$D = \frac{2\pi R\Delta}{360} \qquad 137{,}02 = \frac{2\pi R(32{,}71)}{360} \qquad \Rightarrow R = 240m$$

$$t = R\tan\left(\frac{\Delta}{2}\right) = 240 \times \tan\left(\frac{32{,}71}{2}\right) = 70{,}43m$$

$$b = R\left(\frac{1}{\cos\left(\frac{\Delta}{2}\right)} - 1\right) = 240 \times \left(\frac{1}{\cos\left(\frac{32{,}71}{2}\right)} - 1\right) = 10{,}12m$$

b) By considering that at most half of the centrifugal force can be compensated by the friction, draw the constant centerline superelevation diagram for this curve and show all necessary measurements (km and superelevation values).

$$q_{pra} = 0{,}5 * 0{,}786 * \frac{V^2}{R} = 0{,}5 * 0{,}786 * \frac{80^2}{240} = \%10{,}48$$ $> \%8$ → **Speed limit is required!**

$$8 = 0{,}5 \times 0{,}786 \times \frac{V^2}{240} \Rightarrow V = 69{,}90 km/sa$$

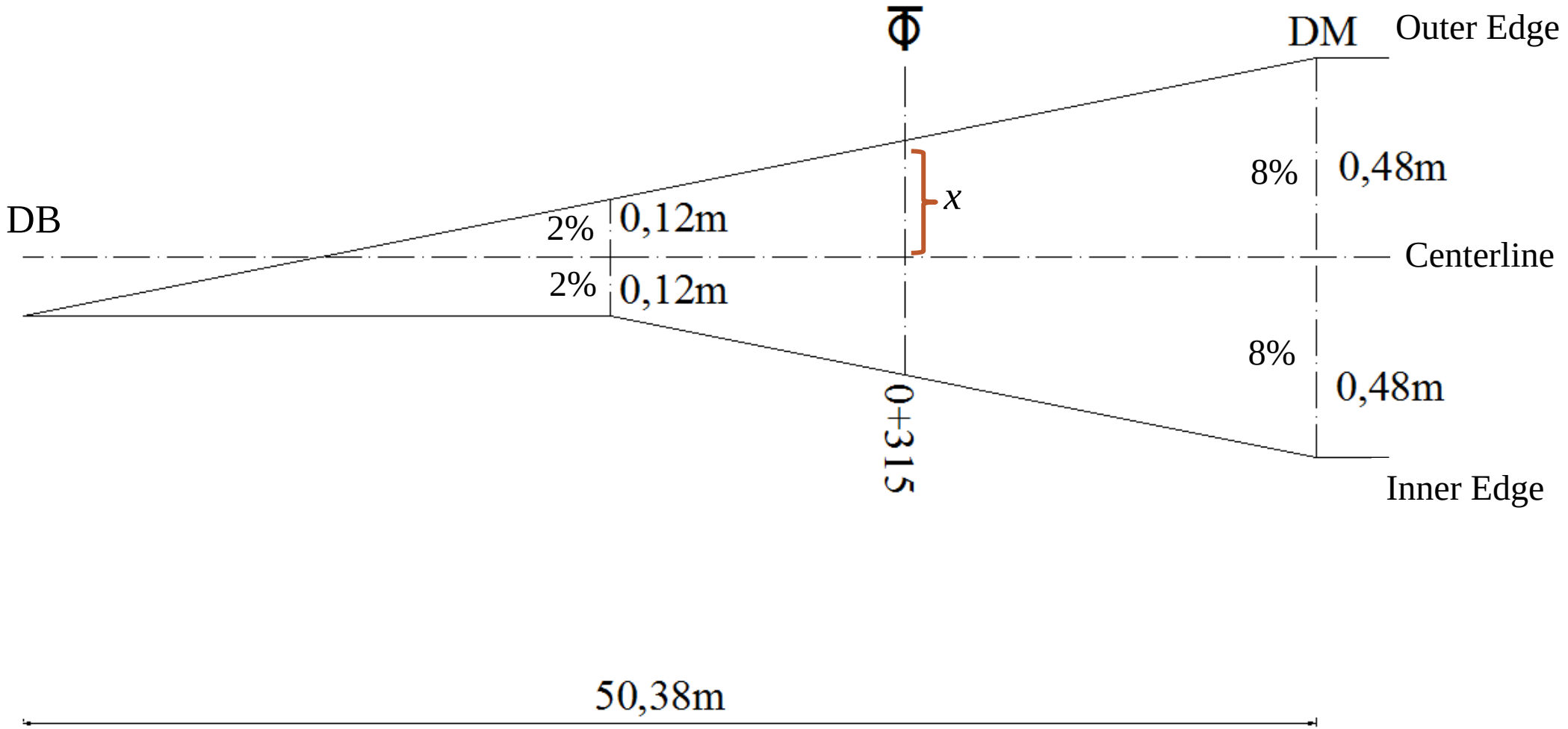
$$L_d = 0{,}0354 \times \frac{V_k^3}{R} = 0{,}0354 \frac{69{,}90^3}{240} = 50{,}38m$$ $> 45m$

$$b = (3{,}5 + 2{,}5) \times 2 = 12\ \text{m}$$

R > 200m no need for expansion.

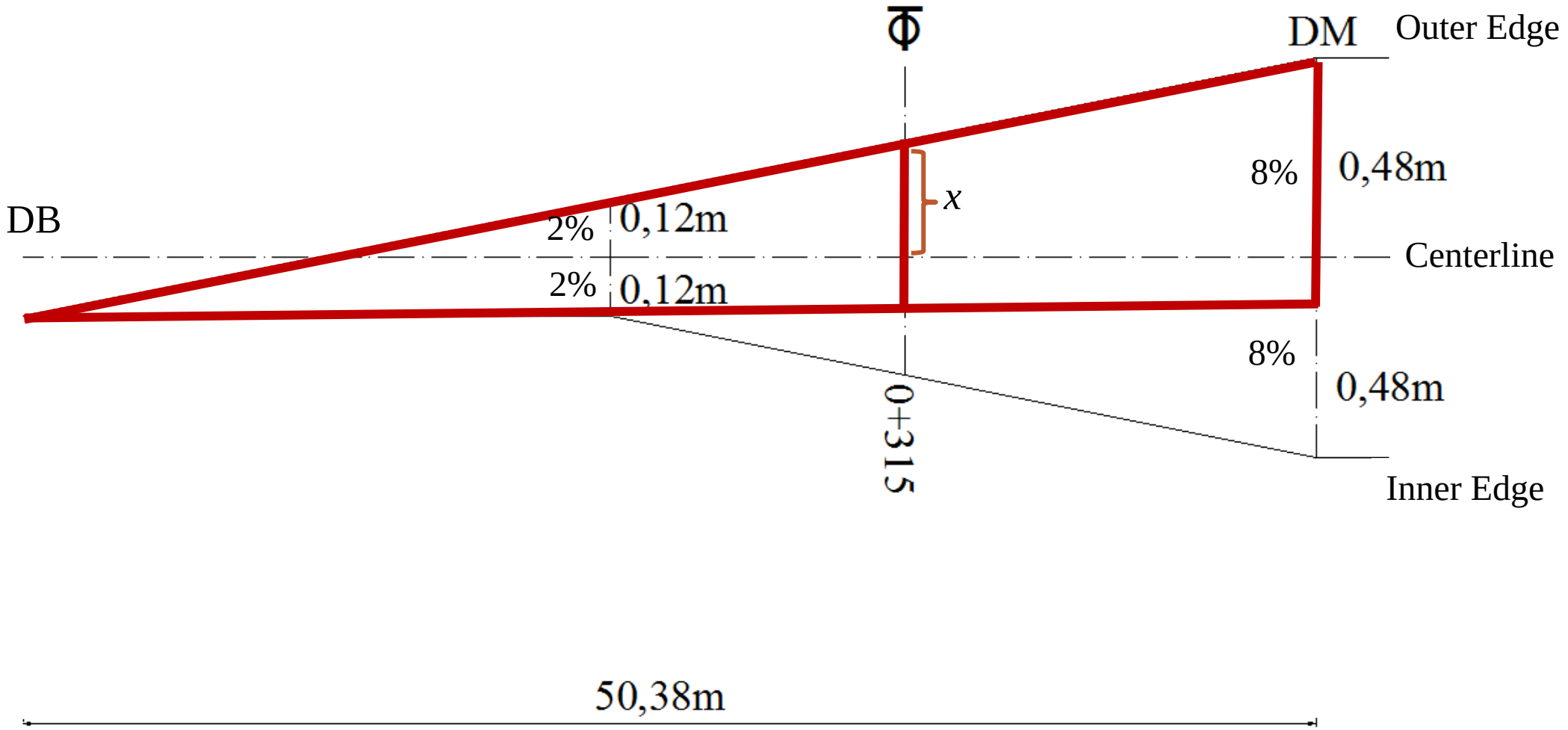
$$h_{\min} = \frac{12}{2} \times 0{,}02 = 0{,}12m$$

$$h_{mals} = \frac{12}{2} \times 0{,}08 = 0{,}48m$$



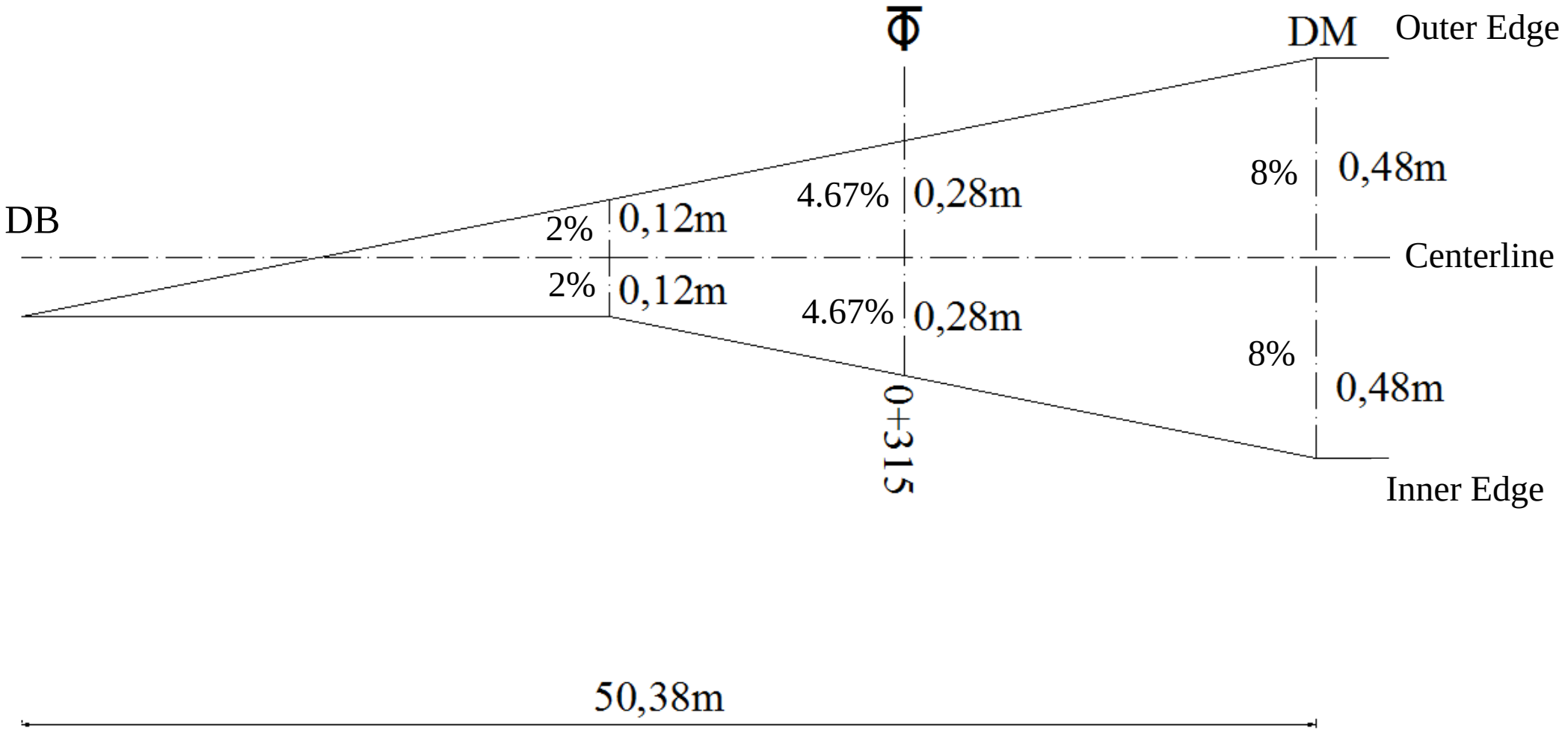
$$DB_{km} = 315 - \frac{2}{3} \times 50,38 = 0 + 281^{41}$$

$$DM_{km} = 315 + \frac{1}{3} \times 50,38 = 0 + 331^{79}$$

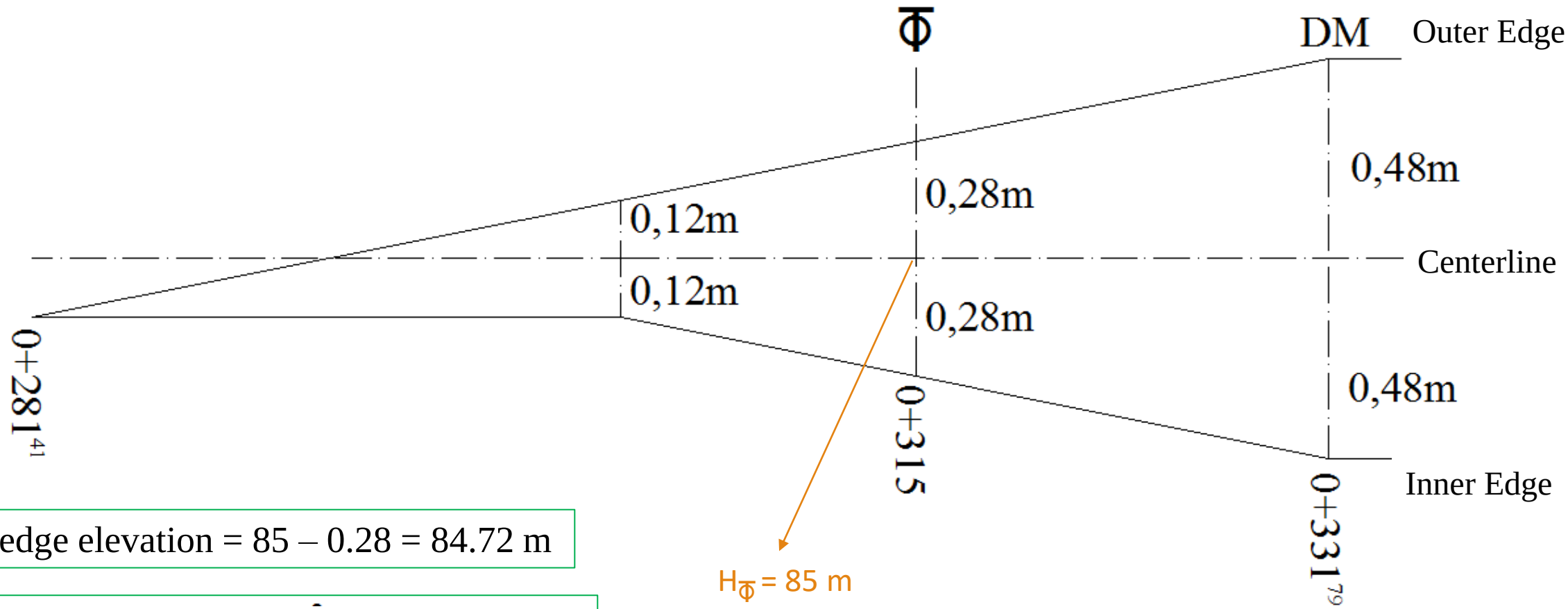


$$\frac{8-(-2)}{50,38}=\frac{x-(-2)}{33,59}\Rightarrow x=\%4,67$$

h_{Φ} = (12/2) x 0,0467 = 0,28m

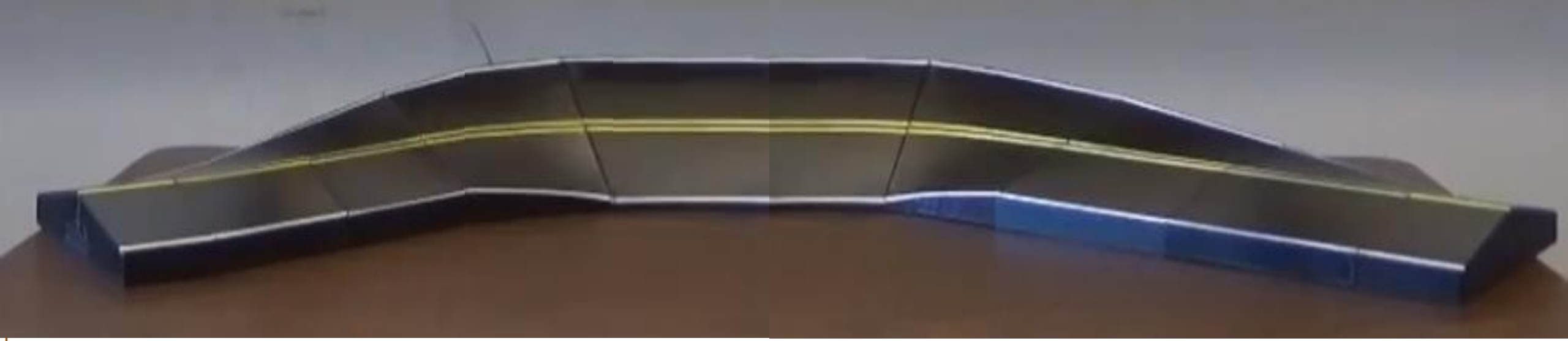

Φ
DM
Outer Edge
DB
2%
0,12m
4.67%
0,28m
8%
0,48m
Centerline
2%
0,12m
4.67%
0,28m
8%
0,48m
0+315
Inner Edge
50,38m

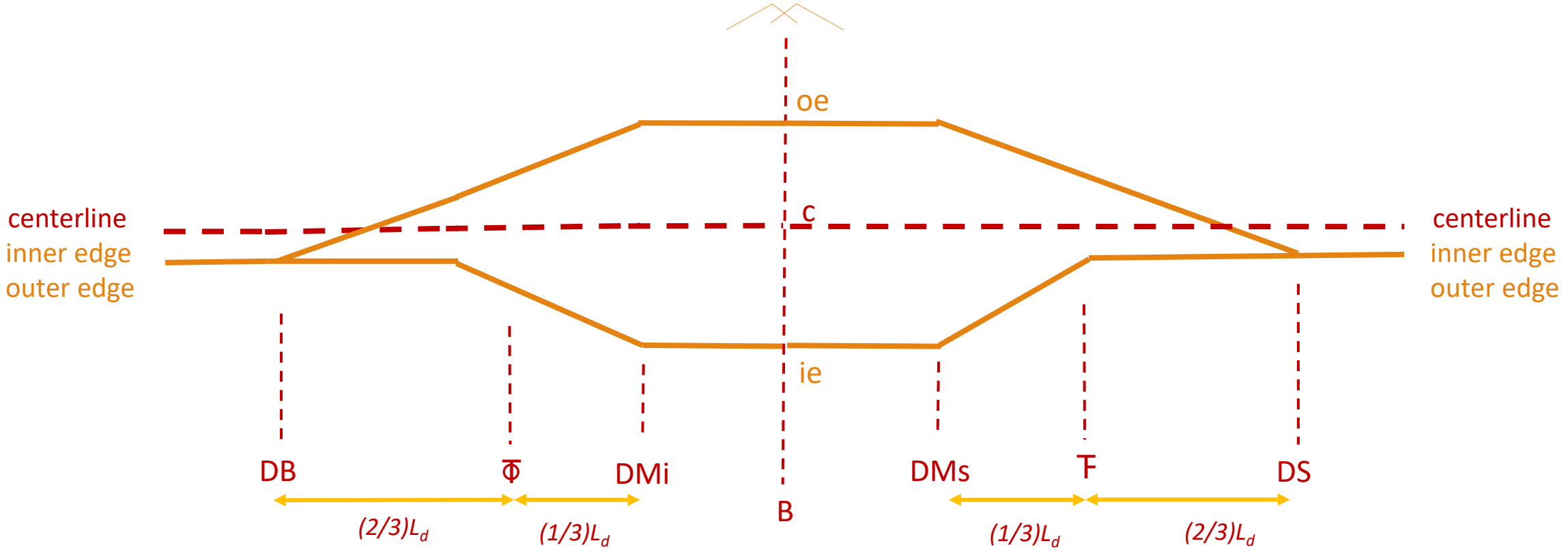
c) The red elevation (grade elevation) of the point Φ is given as 85 m. Calculate the elevations of inner and outer edges.



Inner edge elevation = 85 – 0.28 = 84.72 m

Outer edge elevation = 85 + 0.28 = 85.28 m




oe
centerline
inner edge
outer edge
c
centerline
inner edge
outer edge
ie
DB
Φ
DMi
B
DMs
F
DS
(2/3)L_d
(1/3)L_d
(1/3)L_d
(2/3)L_d

Question – 2

A highway is designed as two-way double lane with a design speed of 90 km/h. Radius of the curve is given as 550 m and beginning point km is given as Φ_{km} = 3+220. It is considered to perform "outer edge constant superelevation" without ramp and without transition curve.

a) Calculate the superelevation application length and draw the superelevation diagram which shows the km values of the critical points.

b) By considering that this route section is 5% upgraded and the centerline elevation of the DB cross section is given as 450 m. Find the red elevations (grade elevations) of the centerline, outer edge and inner edge of Φ cross section.

c) By using the data in part b find the km value and red elevation (grade elevation) of the cross section where the lateral slope (superelevation) is 5%.

Given
Platform width = 10 m
Roof slope = %2 ; Max. superelevation = %8
Theoretical superelevation = 0,786*(V_p^2/R) ; Practical superelevation = 0,443*(V_p^2/R) ; L_d = 0,0354.V_p^3/R

a) Calculate the superelevation application length and draw the superelevation diagram which shows the km values of the critical points.

$$q_{pra} = 0,443 \times \frac{90^2}{550} = \%6,52 \quad < q_{maks} = \%8$$

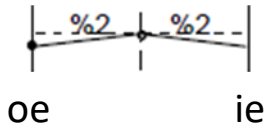
$$L_d = 0,0354 \times \frac{90^3}{550} = 46,92m \quad > L_{\min} = 45m$$

R > 200 m no need for expansion.

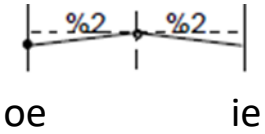
$$h_{\min} = \frac{10}{2} \times 0,02 = 0,10m \qquad h_{maks} = \frac{10}{2} \times 0.0652 = 0,33m$$

$$DB_{km} = \Phi_{km} - ((2/3) * L_d) = (3+220) - (46,92 * 2/3) = 3+188^{72}$$

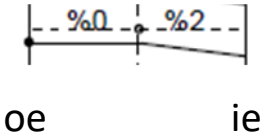
$$DM_{km} = \Phi_{km} + ((1/3) * L_d) = (3+220) + (46,92 / 3) = 3+235^{64}$$



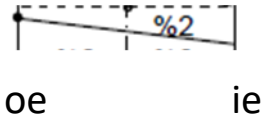
crown slope



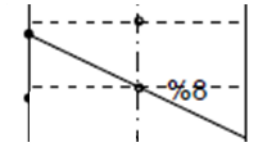
DB



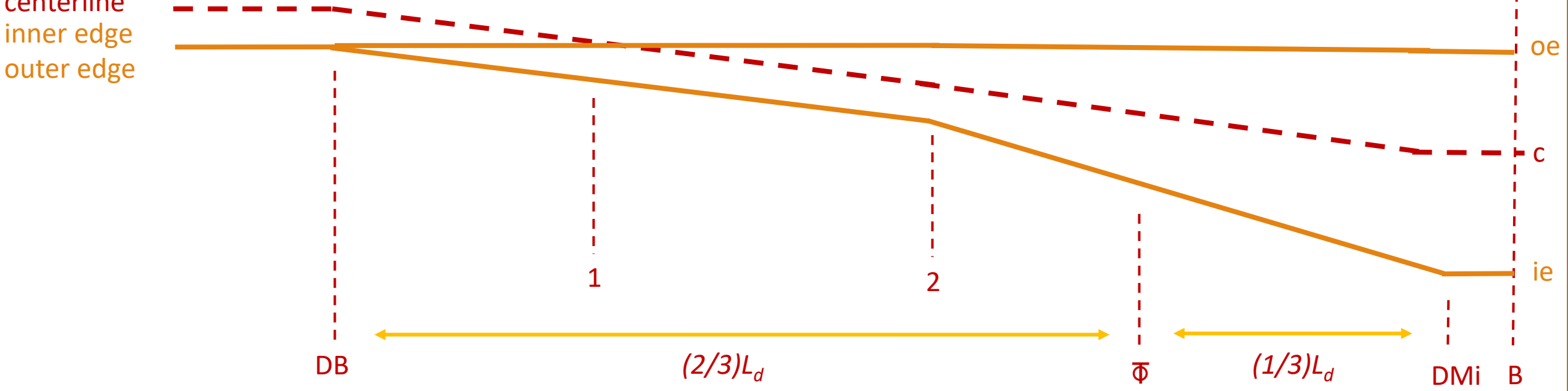
1

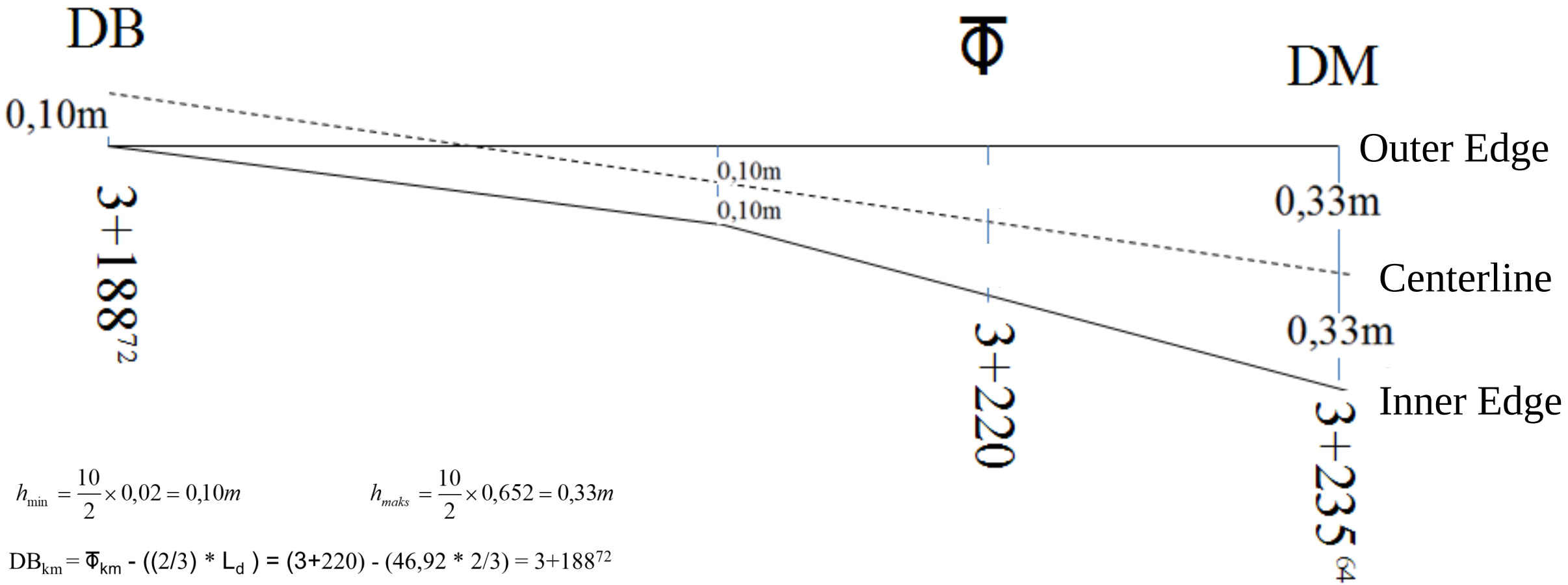


2



DMi





$h_{\min} = \frac{10}{2} \times 0{,}02 = 0{,}10m$ $h_{maks} = \frac{10}{2} \times 0{,}652 = 0{,}33m$

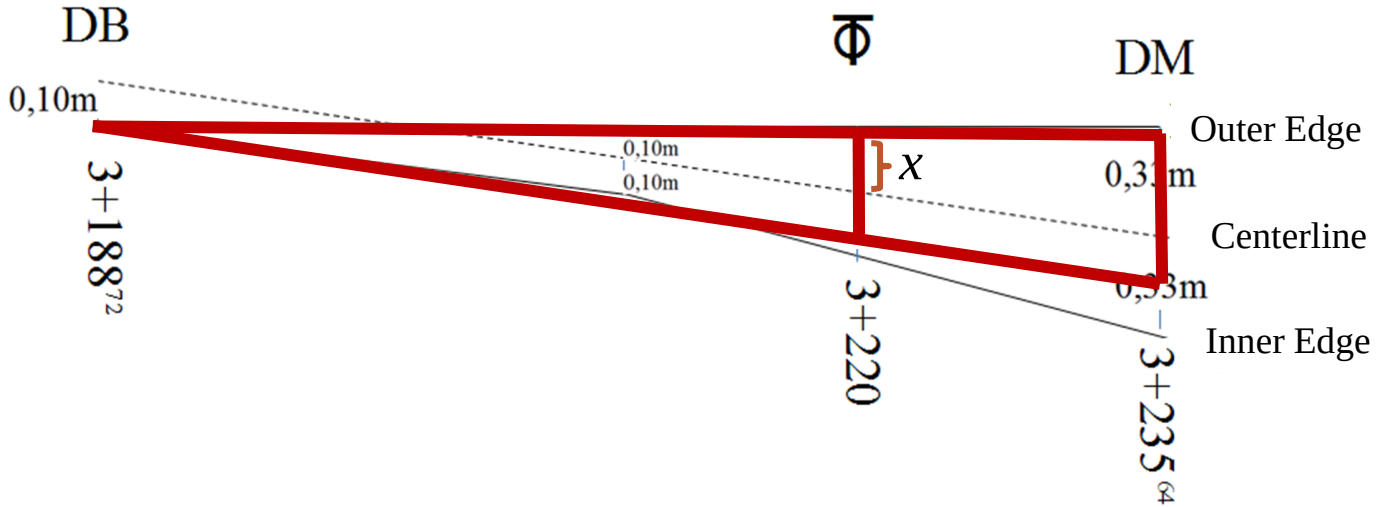
$DB_{km} = \Phi_{km} - ((2/3) * L_d) = (3+220) - (46{,}92 * 2/3) = 3+188^{72}$

$DM_{km} = \Phi_{km} + ((1/3) * L_d) = (3+220) + (46{,}92 / 3) = 3+235^{64}$

b) By considering that this route section is 5% upgraded and the centerline elevation of the DB cross section is given as 450 m. Find the red elevations (grade elevations) of the centerline, outer edge and inner edge of Φ cross section.

superelevation at point Φ: $\frac{6{,}52-(-2)}{46{,}92}=\frac{x-(-2)}{31{,}28}\Rightarrow x=\%3{,}68$

$h_{(\Phi)}$ = 0,0368 * (10/2) = 0,18 m



b) By considering that this route section is 5% upgraded and the centerline elevation of the DB cross section is given as 450 m. Find the red elevations (grade elevations) of the centerline, outer edge and inner edge of Φ cross section.

What would be the elevation at Φ cross section if there was no superelevation?

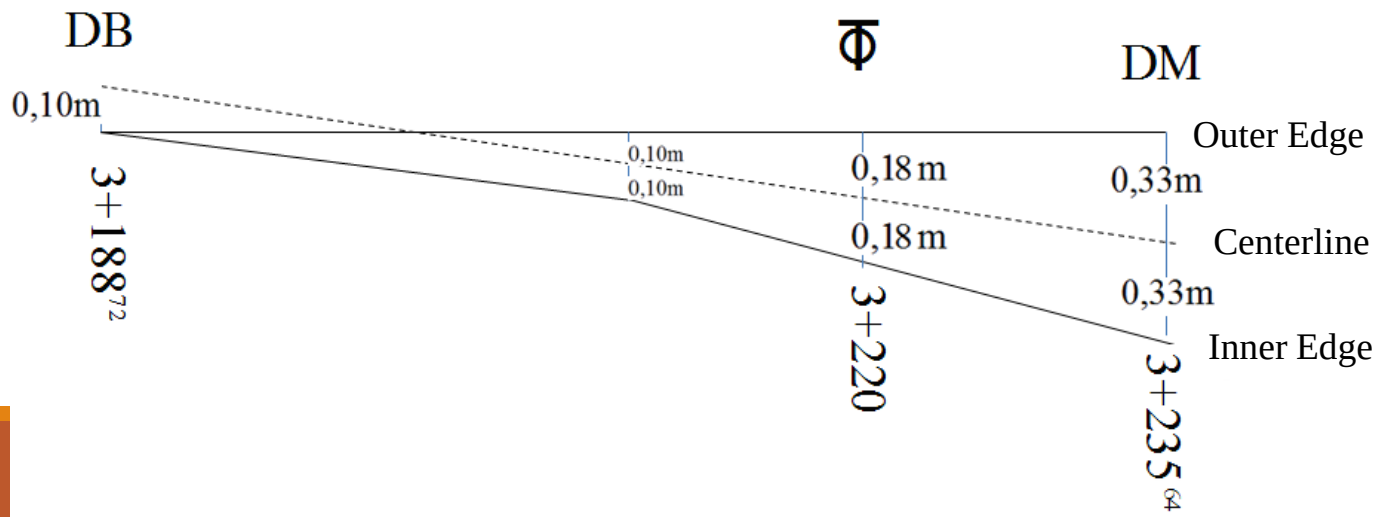
$$H_{(\Phi)} = [450 + (31{,}28 \times 0{,}05)] = 451{,}56m$$

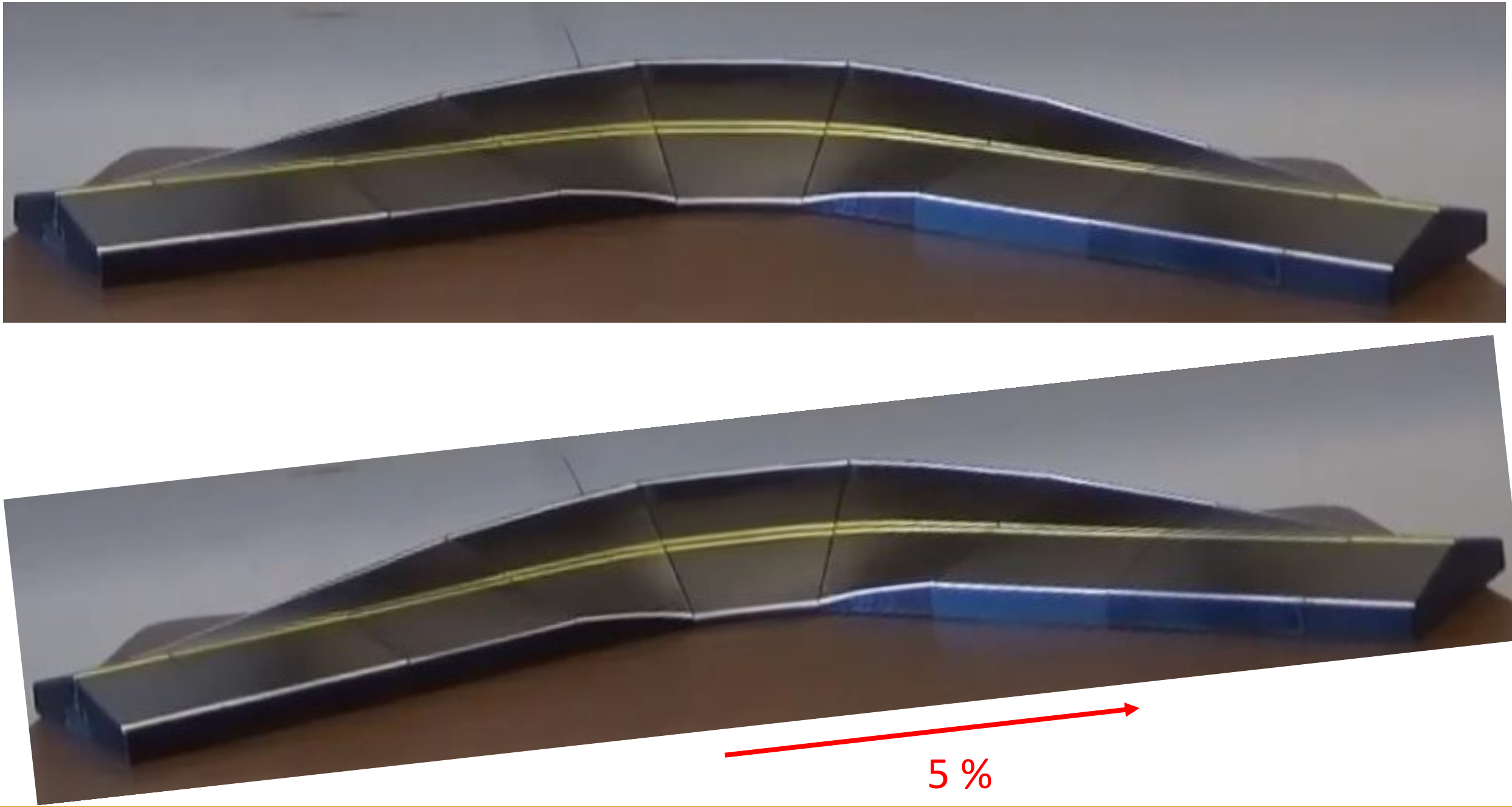
Outer edge is 0.1 m below the original centerline elevation:

$$H_{(\Phi)oe} = 451{,}56\text{-}0{,}10 = 451{,}46 \text{ m}$$

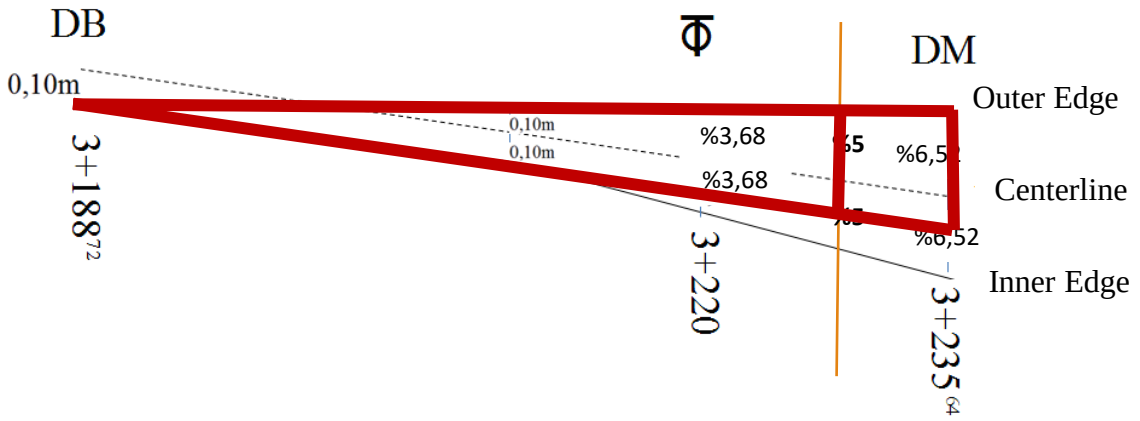
Inner edge is **0,18 x 2 = 0,36 m** lower than the outer edge.

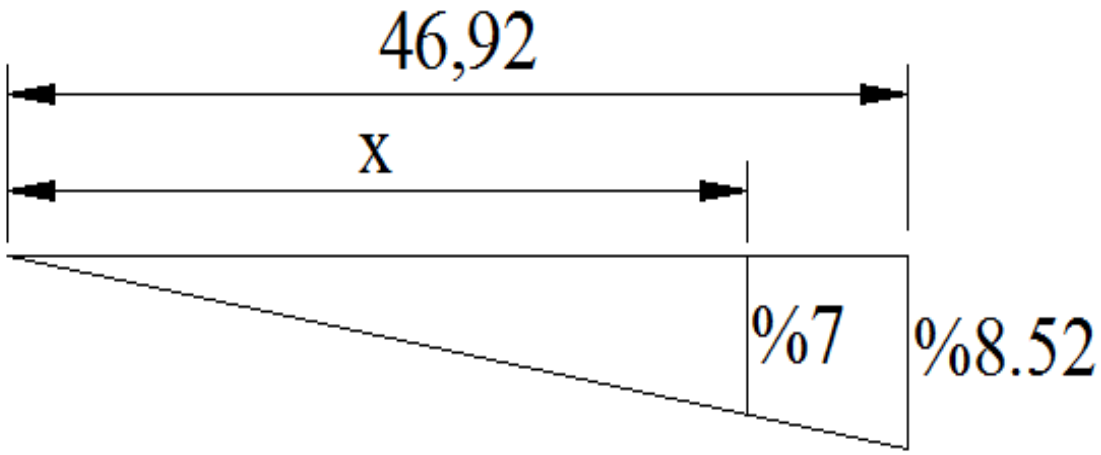
$$H_{(\Phi)ie} = 451{,}46 - (0{,}36) = 451{,}10m$$




5 %

c) By using the data in part b find the km value and red elevation (grade elevation) of the cross section where the lateral slope (superelevation) is 5%.





$$\frac{6,52-(-2)}{46,92}=\frac{5-(-2)}{x}\Rightarrow x=38,55m$$

$$km=(3+188^{72})+38,55=3+227^{27}$$

c) By using the data in part b find the km value and red elevation (grade elevation) of the cross section where the lateral slope (superelevation) is 5%.

What would be the elevation of the centerline at that cross section if there was no superelevation?

$$H = [450 + (38{,}55 \times 0{,}05)] = 451{,}93m$$

Outer edge is 0.1 m below the original centerline elevation.

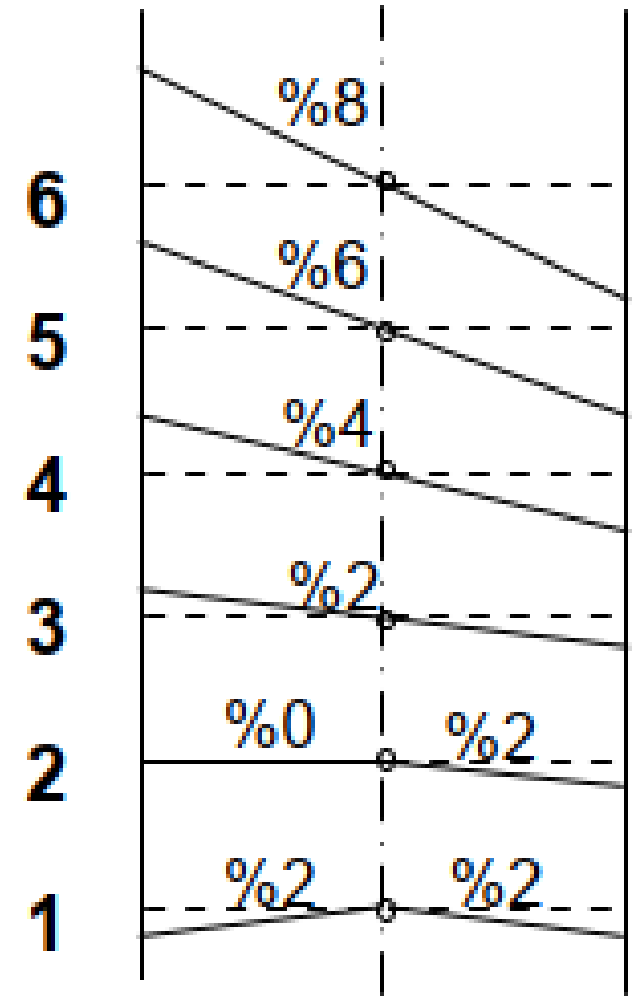
H_{oe} = 451.93 – 0.1 = 451.83 m

Inner edge is 2*(10/2)*0.05 = 0.5 m lower than the outer edge.

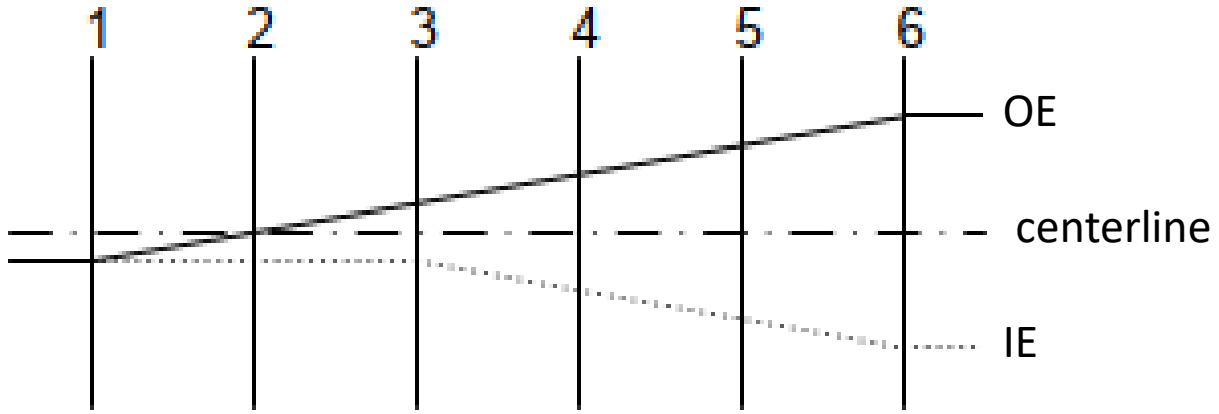
H_{ie} = 451.83 – 0.5 = 451.33 m

CONSTANT CENTERLINE SUPERELEVATION

Cross-section Views

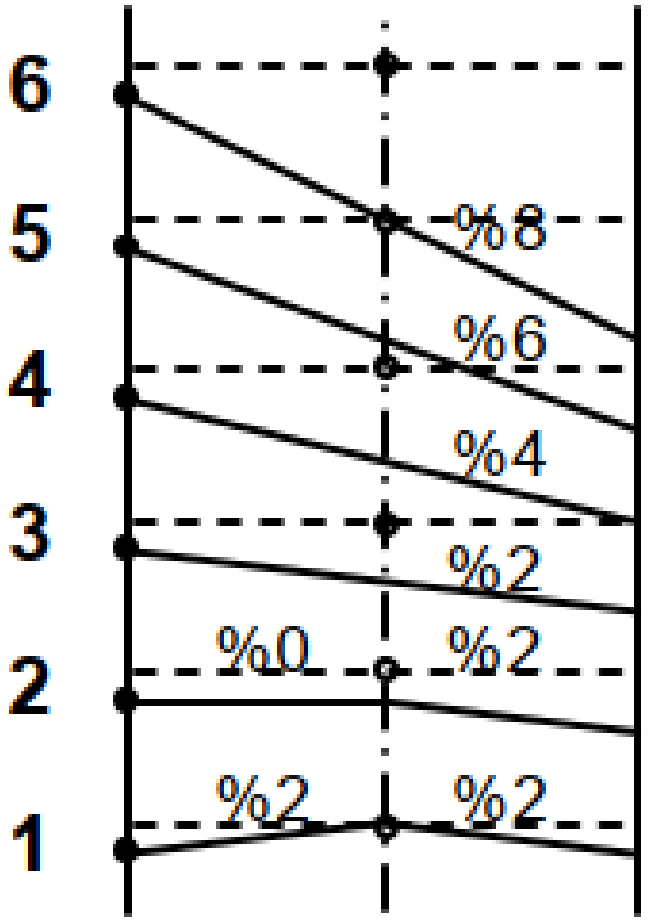


Superelevation Diagram

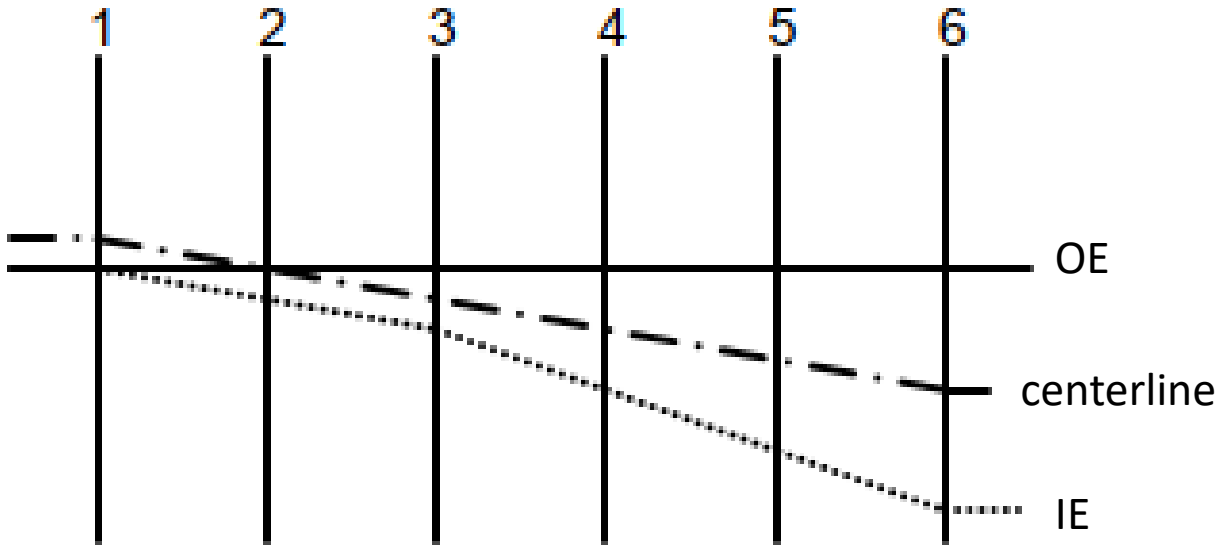


CONSTANT OUTER EDGE SUPERELEVATION

Cross-section Views

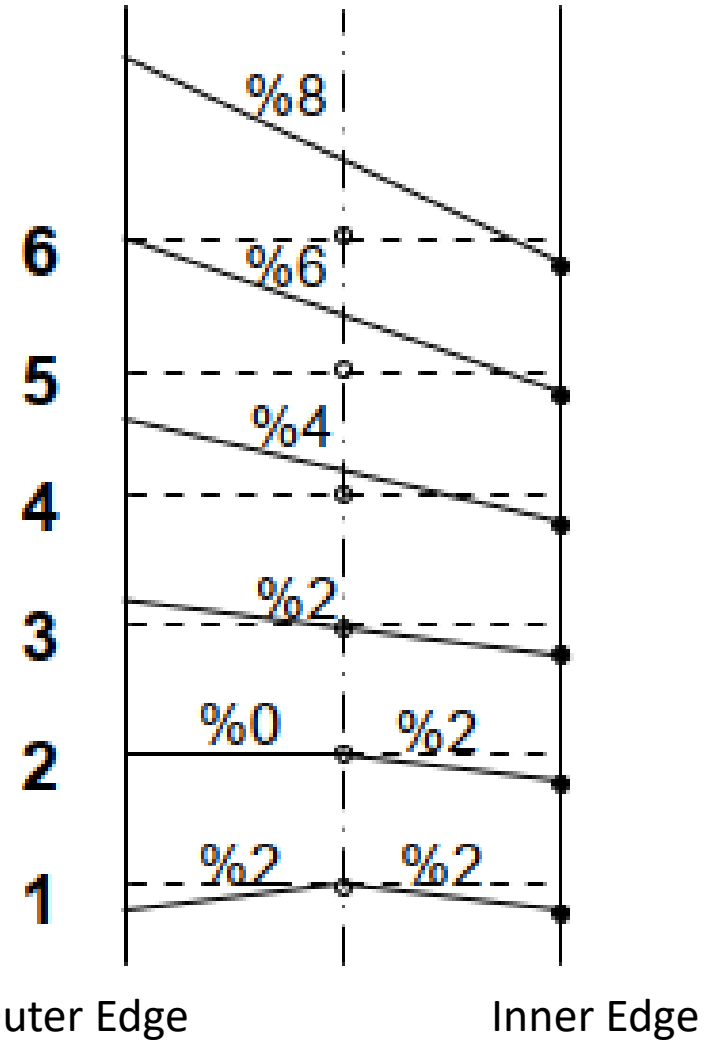


Superelevation Diagram

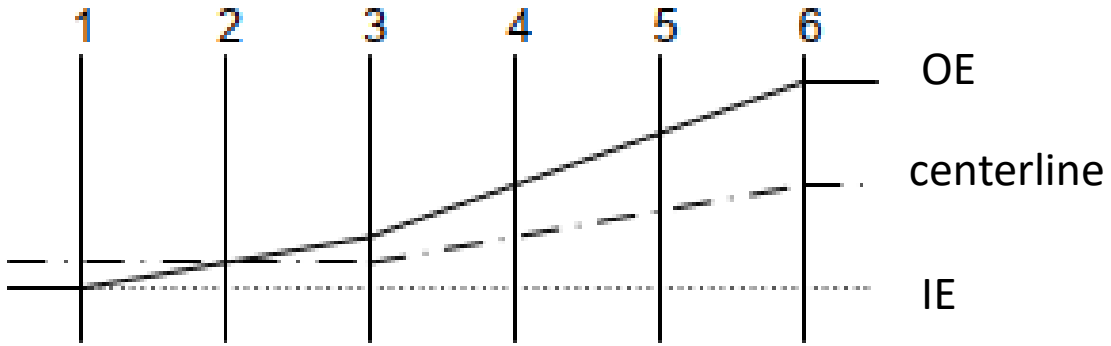


CONSTANT INNER EDGE SUPERELEVATION

Cross-section Views

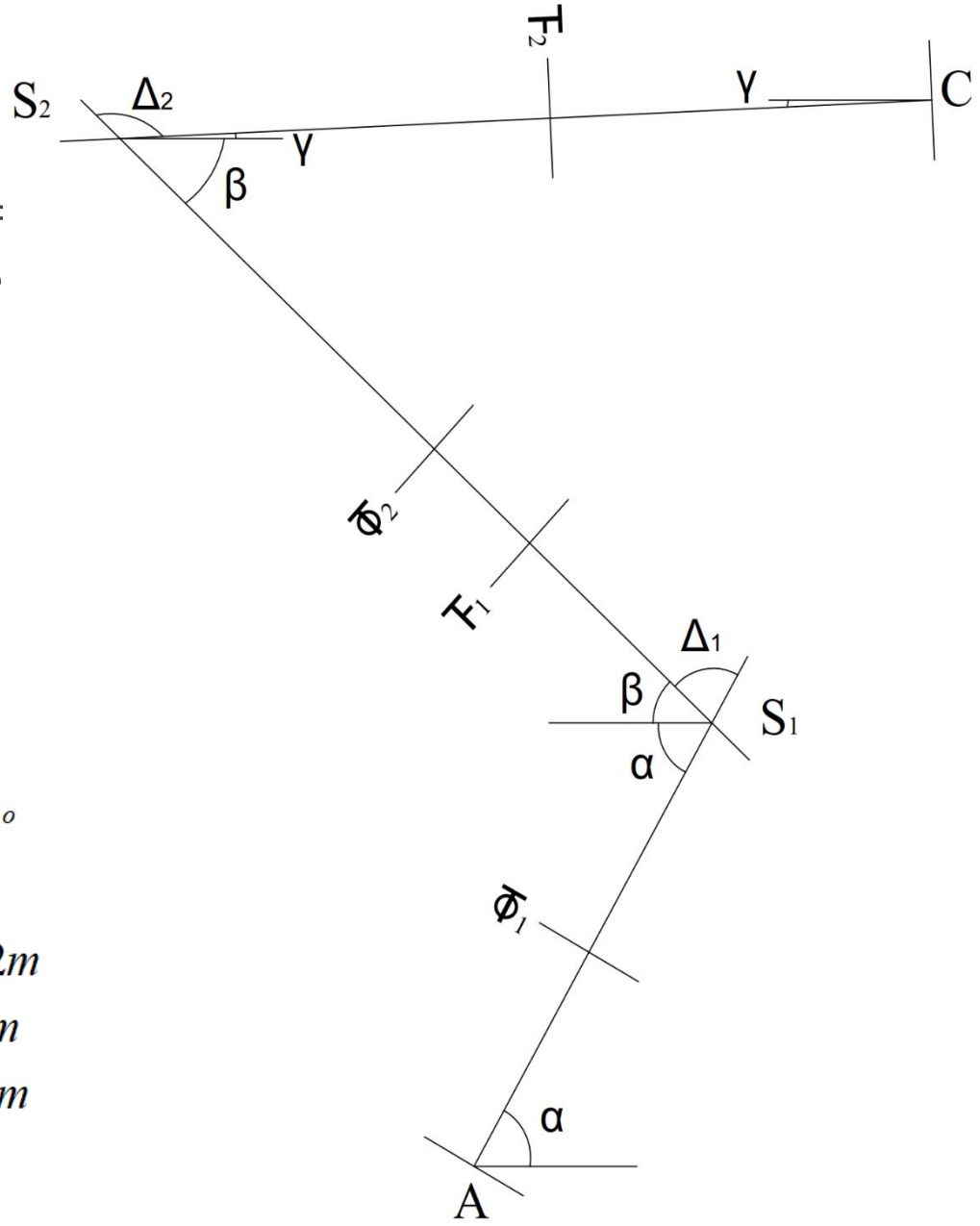


Superelevation Diagram



Question – 3

Perform the precise axis calculation of the route section below. Find the km values for critical points and find the total length of the route. The coordinates of the critical points and the curve components determined by the route study are provided below.



Point	X	Y
A	693	802,5
S1	1064	1012,5
S2	1553	490
C	1585	1205

Route Study results

1.Curve

$\Delta_1 = 75{,}69^o$

$R_1 = 280m$

$D_1 = 369{,}89m$

$t_1 = 217{,}54m$

$b_1 = 74{,}58m$

2. Curve

$\Delta_2 = 133{,}90^o$

$R_2 = 160m$

$D_2 = 373{,}92m$

$t_2 = 376{,}02m$

$b_2 = 248{,}65m$

Point	X	Y	ΔX	ΔY	Distance
A	693	802,5			
S1	1064	1012,5	371	210	426,31
S2	1553	490	489	522,5	715,63
C	1585	1205	32	715	715,72

$$tg\alpha = \frac{371}{210} \Rightarrow \alpha = (60,49)^o$$

$$tg\beta = \frac{489}{522,5} \Rightarrow \beta = (43,10)^o$$

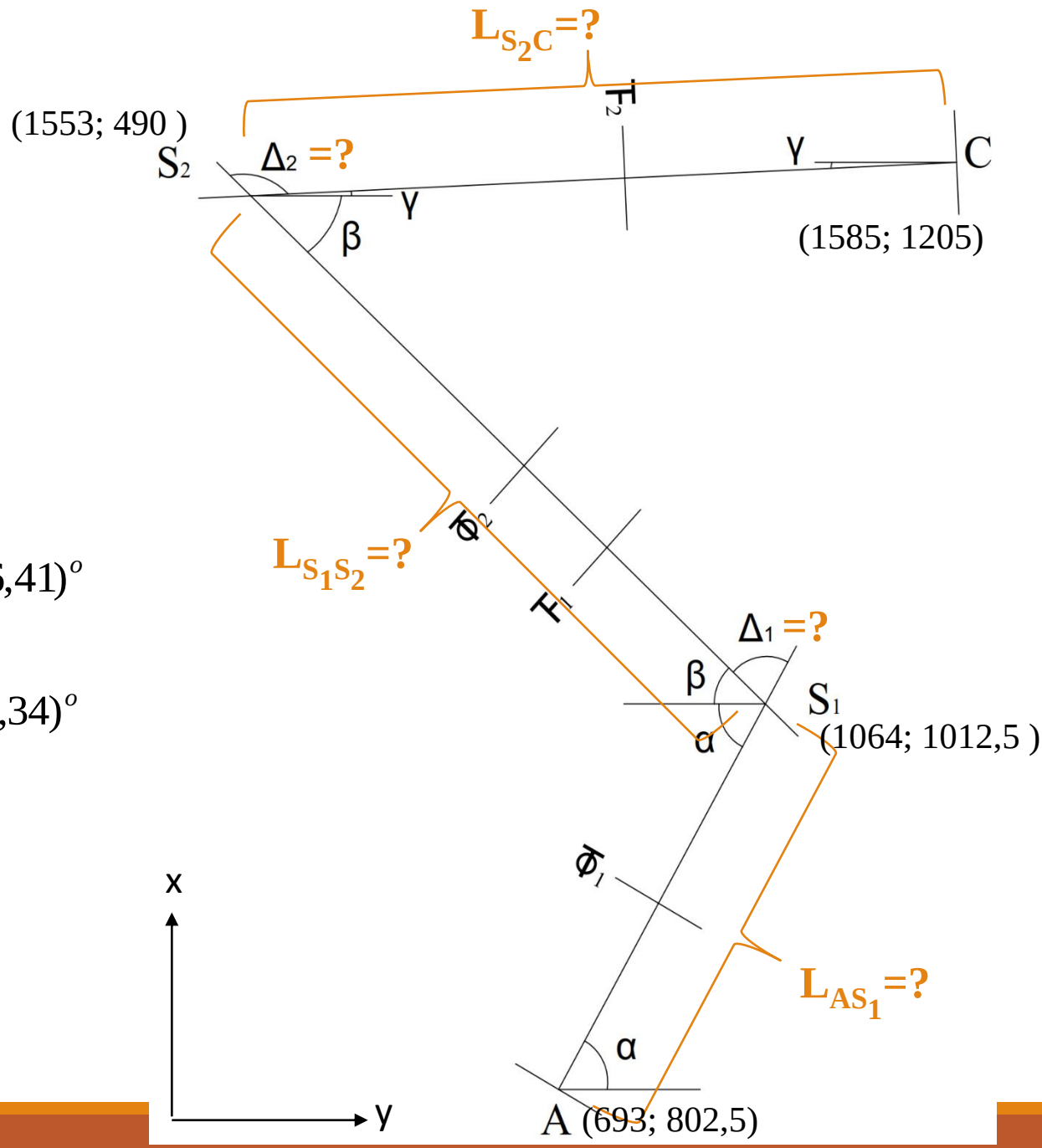
$$tg\gamma = \frac{32}{715} \Rightarrow \gamma = (2,56)^o$$

$$\Delta_1 = 180 - (\alpha + \beta) = (76,41)^o$$

$$\Delta_2 = 180 - (\gamma + \beta) = (134,34)^o$$

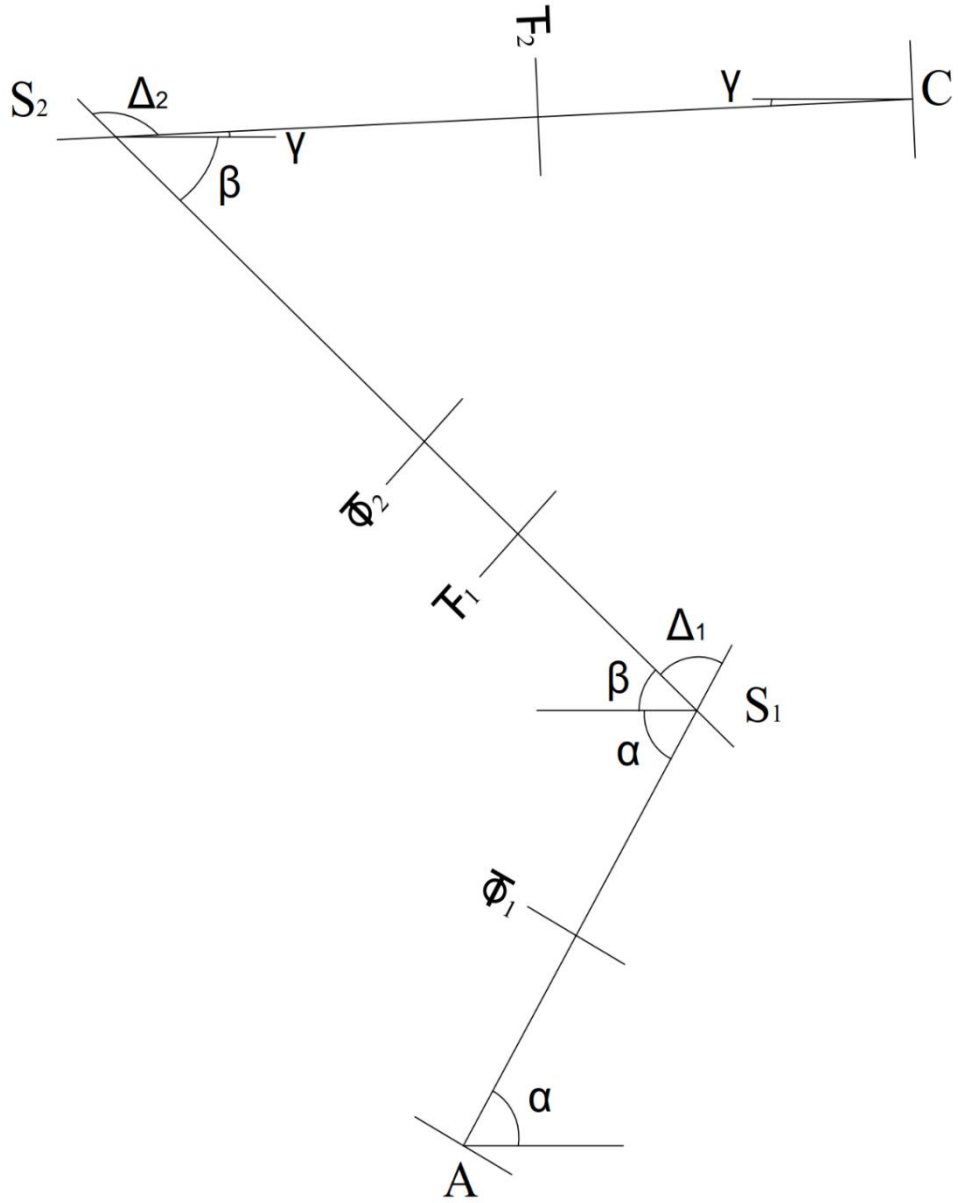
$$\overline{AS_1} = \sqrt{\Delta X^2 + \Delta Y^2} = \sqrt{(371)^2 + (210)^2} = 426,31m$$

$$\overline{S_1S_2} = 715,63m \qquad \overline{S_1C} = 715,72m$$



Route Study results **Precise Values**

1.Curve	2. Curve	1. Curve	2. Curve
$\Delta_1 = 75{,}69^o$	$\Delta_2 = 133{,}90^o$	$\Delta_1 = 76{,}41^o$	$\Delta_2 = 134{,}34^o$
$R_1 = 280m$	$R_2 = 160m$	$R_1 = 280m$	$R_2 = 160m$
$D_1 = 369{,}89m$	$D_2 = 373{,}92m$	$D_1 = 373{,}41m$	$D_2 = 375{,}15m$
$t_1 = 217{,}54m$	$t_2 = 376{,}02m$	$t_1 = 220{,}38m$	$t_2 = 380{,}07m$
$b_1 = 74{,}58m$	$b_2 = 248{,}65m$	$b_1 = 76{,}32m$	$b_2 = 252{,}37m$



Calculation of the Km values for critical points by using precise calculations

$\Phi_1(\text{km}) = A(km) + (\overline{AS_1} - t_1) = 0 + 205^{93}$

$B_1(\text{km}) = \Phi_1(\text{km}) + (D_1/2) = 0{+}392^{64}$

$F_1(\text{km}) = \Phi_1(\text{km}) + D_1 = 0{+}579^{34}$

$\Phi_2(\text{km}) = F_1(\text{km}) + (\overline{S_1S_2} - t_1 - t_2) = 0 + 694^{52}$

$B_2(\text{km}) = \Phi_2(\text{km}) + (D_2/2) = 0{+}882^{10}$

$F_2(\text{km}) = \Phi_2(\text{km}) + D_2 = 1{+}069^{67}$

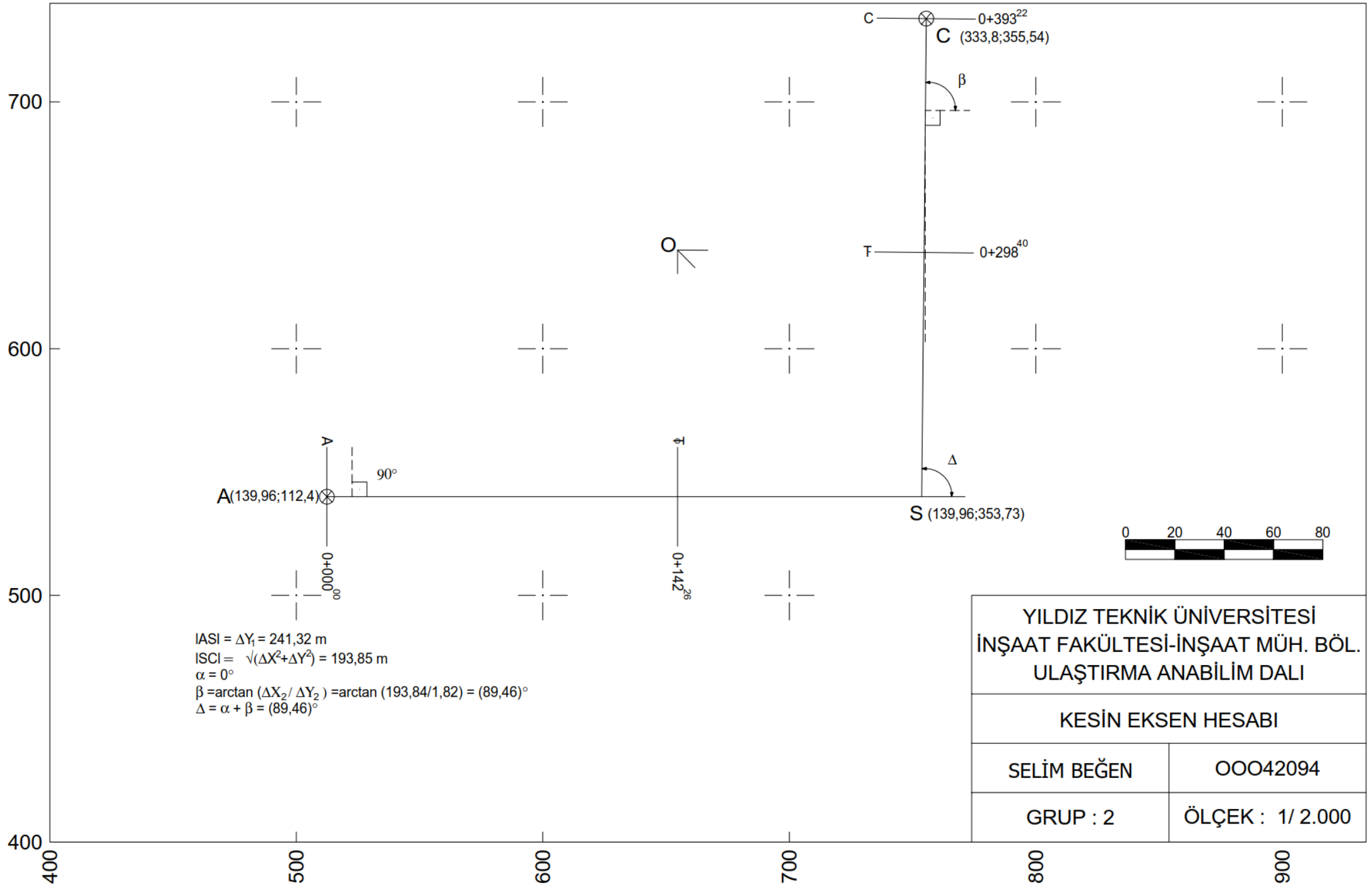
$C(\text{km}) = F_2(\text{km}) + (\overline{S_2C} - t_2) = 1 + 405^{32}$

PRECISE AXIS CALCULATIONS

The length of the route calculated during route study is an approximate value. Therefore the plan obtained is not precise but temporary and if it is necessary some arrangements can be performed. In order to make the study more realistic precise axis study will be performed. Precise axis calculation is calculation of the route length by using analytical geometry.

Prepare the precise axis calculation sheet by transferring the alignments on the route study sheet to the map without grid and without contour lines and without the arrow directed North. Close unnecessary layers. Read the beginning and ending km of the route and km of the intersection point. The calculations should be performed in meters as unit. On the coordinate axis Y is the horizontal and X is the vertical. According to that the coordinates of the critical points should be written.

Also, the angles between that the alignments and the horizontal and vertical axis and curve angles should be shown symbolically. The values of the angles and length of the alignments should be written on the map somewhere empty. The critical cross sections and km values (beginning and end points of the curve and alignment) should be marked on alignments. Line weights should be similar to the route study sheet. Than precise plan drawing can be performed.



IASI = ΔY_1 = 241,32 m

ISCI = $\sqrt{(\Delta X^2+\Delta Y^2)}$ = 193,85 m

$\alpha = 0°$

β =arctan ($\Delta X_2 / \Delta Y_2$) =arctan (193,84/1,82) = (89,46)°

$\Delta = \alpha + \beta$ = (89,46)°

YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
KESİN EKSEN HESABI	
SELİM BEĞEN	OOO42094
GRUP : 2	ÖLÇEK : 1/ 2.000